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Prescribed-time stabilisation of nonlinear time-varying systems exhibiting impulsive behaviour

Arnab Mapui and Santwana Mukhopadhyay

Department of Mathematical Sciences, Indian Institute of Technology (BHU), Varanasi, India

ABSTRACT

This article investigates the prescribed-time stabilisation of time-varying impulsive nonlinear systems. This work's principal contributions are twofold. In the first part, in contrast to the traditional approaches to prescribed-time stability, where continuous systems used to be considered globally stable, it is shown here that even the systems with unstable flows can be stabilised globally in prescribed time by properly establishing relations between time-varying gain functions, stabilising impulsive sequences, and the prescribed time. Some adequate results on these aspects are proposed. An analysis of how destabilising impulses affect the stability of time-varying systems is presented in the second part of the article. Some results on local prescribed-time stability under the influence of destabilising impulses are proposed in this respect. Additionally, the suggested findings are used to the construction of prescribed-time controllers for time-varying affine systems that have input disturbances. To further support the veracity of the suggested results, three examples are given.

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control

1. Introduction

Nowadays, researchers are mainly focused on dynamical systems that satisfy the system's objective in some finite time. Unlike asymptotic or exponential stability, which requires an infinite duration, it can guarantee the convergence of state trajectories to the intended location in a finite amount of time. In this aspect, pioneering work on finite-time stability (FTS) can be found in Bhat and Bernstein (2000), Haimo (1986), and Roxin (1966). After the initiation of this new notion of stability, it has been explored in different fields such as robotics (Galicki, 2015; Pliego-Jiménez et al., 2020), secure communication (Perruquetti et al., 2008), power systems (Zhao et al., 2011), Neural Networks (Jiang et al., 2015), and many more. Though, FTS is superior to asymptotic/exponential stability in terms of faster finite time convergence and robustness against disturbances, one of its shortcomings is that the settling time is dependent on the states' initial conditions. To address this problem, Polyakov (2011) introduced the notion of fixed-time stability (F_XTS). Here, an upper bound of the settling time is introduced, which is independent of the initial conditions. Instead, it depends on the parameters of the system. Some noteworthy contributions on F_XTS can be found in Hu et al. (2017), Yu et al. (2019), Moulay and Léchappé (2021), Jamal et al. (2022), and Mapui et al. (2024a). Though the upper bound of settling time is independent of initial conditions, it cannot be assigned in advance, as it is needed in many practical applications in the likes of missile guidance (Jing et al., 2021), robot control (Munoz-Vazquez et al., 2019). Hence, the term predefined-time stability (P_DTS) was coined in Sánchez-Torres et al. (2014). Here, the user can assign the upper bound of settling time

in advance depending on the system's tunable parameters. Some prominent work in this regard can be found in Aldana-López et al. (2019), Wei and Li (2023), Jiménez-Rodríguez et al. (2020), and Mapui et al. (2024b). For instance, Aldana-López et al. (2019) provide a sufficient Lyapunov condition based on the polynomial form as provided in Polyakov (2011); whereas, in Wei and Li (2023), authors have discussed P_DTS of impulsive system using hybrid impulses. Moreover, Jiménez-Rodríguez et al. (2020) came up with a novel Lyapunov form based on a special kind of function that generalises many previously discussed Lyapunov sufficient conditions on autonomous systems. Despite remarkable contributions, P_DTS suffers from the drawback of requirement for tunable parameters of the system. Thus, a new form of stability, prescribed-time stability (P_STS), has been introduced in the community of researchers of stability properties of dynamical systems through the prodigious work reported by Song et al. (2017). The main characteristic of P_STS is that the upper bound of settling time is independent of the initial conditions as well as of the system parameters and can be pre-assigned by the user. Several works in this respect can be found in Hua et al. (2021), Ning et al. (2023), Orlov (2022), Zhou and Zhang (2022), and Zhou and Shi (2021). Specifically, Hua et al. (2021) provide a P_STS theorem for uncertain nonlinear systems and is applied to prescribed-time convergence of high-order nonlinear systems via the backstepping method. Whereas Ning et al. (2023) study prescribed-time control of a class of uncertain nonlinear time-delay systems. The author of Orlov (2022) concentrates on a novel idea of squeezing the infinite interval into the prescribed one by suitable time deformation. A linear time-varying approach for P_STS is

proposed by Zhou and Zhang (2022), which works as a unified framework for many finite and fixed-time stability theorems. Moreover, Zhou and Shi (2021) focus on the $P_S TS$ problem of unknown nonlinear functions satisfying specific linear growth properties. Recently, in Xu and Liu (2023), the authors have mentioned that, in most of the articles on $P_S TS$, the time-varying gain function used to go to infinity as time approaches the prescribed time, which makes the right-hand side of the Lyapunov inequality infinity and thus has no sense in real applications. Hence, it is essential to check the validity of $\dot{V}_L(T_L) = 0$, where V_L is the Lyapunov function and T_L is the prescribed time. This validation is missing in most of the work reported in the literature on $P_S TS$. In view of the above, our first motivation is to provide an inevitable Lyapunov inequality for prescribed-time stability of nonlinear impulsive dynamical systems which consider the interval $[0, T_L]$ rather than $[0, T_L)$ as used in most of the articles and also to use time-varying gain functions whose integration over the interval $[0, T_L)$ is finite instead of being infinite.

Meanwhile, being a class of hybrid systems, impulsive systems have become a fascinating topic among researchers due to their ability to describe complex dynamical systems involving continuous and discrete parts. The evolution of the state trajectories is taken care of by the continuous part. At the same time, the discrete part denotes any changes at discrete points. Some notable works on FTS , $F_X TS$ and $P_D TS$ of impulsive dynamical systems can be found in Wei and Li (2023), Li et al. (2019), Xing et al. (2023), and Nersesov and Haddad (2008). For instance, Li et al. (2019) discuss some Lyapunov sufficient conditions for settling-time estimation of nonlinear impulsive systems depending on the effects of impulses (both stabilising and destabilising). In Xing et al. (2023), the FTS of nonlinear impulsive systems under the effects of vanishing and non-vanishing disturbances is studied. At the same time, the authors in Wei and Li (2023) have provided sufficient conditions for $F_X TS$ of impulsive systems constituting destabilising and hybrid impulses without linear feedback regulation. They have also provided $P_D TS$ under the effect of hybrid impulses with cumulative stabilising effects. Recently, Li et al. (2023) have given a new $P_S TS$ theorem for nonlinear impulsive piecewise-smooth systems and is applied to Chua's circuit.

It should be noted that most of the aforementioned articles and their citations use stable flows of the systems. So, generally, the question arises of whether impulses can be regulated in certain ways to make an unstable system stable. In the answer to this, two of the major contributing articles are He et al. (2022) and Xing et al. (2024). In He et al. (2022), stabilising impulses are regulated not only to stabilise nonlinear autonomous impulsive systems with unstable flows but also to stabilise within a prescribed settling time. However, Xing et al. (2024) enlightened the researchers of this particular field by providing some results to stabilise nonlinear, non-autonomous impulsive systems with unstable flows in finite time with the help of a proper combination of time-varying gain function, systems structure and placement of stabilising impulsive sequences. Here, authors have further shown the effect of destabilising impulses on the stability behaviour of the systems. Though both the articles He et al. (2022) and Xing et al. (2024) have provided a considerable amount of valuable information, there still exist much

scope to work with. For example, the prescribed-time stability in He et al. (2022) is local. Whereas the integration of the time-varying gain function in the article Xing et al. (2024) tends to infinity as the time tends to the finite time, which may reduce the practical applicability. Nevertheless, can one provide global prescribed-time stability results for nonlinear non-autonomous impulsive systems with unstable flows by considering more practical time-varying gain functions and constructing corresponding impulsive sequences remains unrevealed! This is the second motivation for us.

Based on the above discussions, the main contributions of the article can be summarised as follows:

- (1) Compared to most of the recent articles (such as Orlov, 2022; Zhou & Shi, 2021; Zhou & Zhang, 2022), which studied prescribed-time stability of nonlinear impulsive dynamical systems over the interval $[0, T_L)$, we have provided a prescribed-time stability result for nonlinear, time-varying impulsive systems over the interval $[0, T_L]$ using a proper combination of time-varying gain functions, structural characteristics of the system, and stabilising impulses; see Theorem 3.1.
- (2) In contrast to the article (He et al., 2022), which studied local prescribed-time stability, we have studied global prescribed-time stability via Theorem 3.1. Further the finite-time stability result of Xing et al. (2024) has been improved to prescribed-time stability. Also the requirement of the infinite value of the integration of the time-varying gain function of the articles Orlov (2022) and Xing et al. (2024) has been relaxed to a finite one.
- (3) A thorough investigation of the effect of destabilising impulses on prescribed-time stability of impulsive systems is performed over a local region with stable flows, see Theorem 3.2.
- (4) As an application part, we have considered time-varying affine systems as they constitute several practical examples such as Robotic Arms, Spacecraft and Car stabilisation, Electrical Circuits, etc. and, the controllers are designed, in these aspects, to achieve prescribed-time stability of time-varying affine systems involving input disturbances; refer to Theorem 4.1.
- (5) Four instances are created to demonstrate the rationale and veracity of the suggested outcomes, consider Section 5.

The rest of this article is organised as follows. Some important preliminaries are mentioned in Section 2. Section 3 constitutes the main results of the article. Section 4 is dedicated to the application part. Numerical simulations to verify the proposed results are provided in Section 5. Finally, Section 6 concludes this article with major highlights arising out of present investigation and the scope for future work.

2. Preliminaries

2.1 Notations

A variety of notations, system's description, definitions, and an example are included in this section, which serve as a preliminaries to the rest of this article.

- $\mathbb{R}$ is the set of real numbers; $\mathbb{R}_+ = \{z \in \mathbb{R} : z > 0\}$; $\mathbb{R}_{\geq 0} = \mathbb{R}_+ \cup \{0\}$; $\bar{\mathbb{R}}_+ = \mathbb{R}_+ \cup \{\infty\}$.
- $\mathbb{R}^n$ is the n -dimensional real space with the Euclidean norm $|\cdot|$.
- $\mathbb{N}$ is the set of positive integers.
- $\text{sgn}(y)$ represents the signum function for $y \in \mathbb{R}$.
- For any matrix A , A^T denotes the transpose of A .
- For $y = (y_1 \cdots y_n)^T \in \mathbb{R}^n$, $[\text{sgn}(y)]^T = [\text{sgn}(y_1) \cdots \text{sgn}(y_n)]^T$.

2.2 System's description

Let us investigate the subsequent nonlinear impulsive system that evolves with time:

$$\begin{cases} \dot{y}(t) = f(t, y(t)), & t \geq 0, t \neq t_s, \\ y(t) = g(t^-, y(t^-)), & t = t_s, \\ y(0) = y_0, \end{cases} \quad (1)$$

where $y \in \mathbb{R}^n$ is the state vector; $\dot{y}(t)$ is the right-hand derivative of $y(t)$; f, g are the continuous functions from $\mathbb{R}_{\geq 0} \times \mathbb{R}^n$ to $\mathbb{R}^n$ satisfying $f(t, 0) = g(t, 0) = 0$ for all $t \geq 0$; $\{t_s, s \in \mathbb{N}\}$ (in short $\{t_s\}$) is either finite or infinite impulsive sequences satisfying $0 < t_1 < t_2 < \cdots < t_s \rightarrow +\infty$ as $s \rightarrow +\infty$. Let Λ denotes such a class of impulsive sequences. It is also assumed that f and g fulfil conditions that ensure the unique existence of the solution $y(t) = y(t, 0, y_0)$ of the system (1) in forward time (see T. Yang, 2001). In addition, without losing generality, it is assumed that the solution $y(t)$ is right continuous and has left limits at all times.

2.3 Some definitions

Definition 2.1: For any $T_L > 0$, a prescribed constant, the origin of system (1) is said to be uniformly locally P_S TS with respect to T_L over the class of impulsive sequence Λ , if

- for any $\varepsilon > 0$ and $\{t_s\} \in \Lambda$, there exists $\gamma(\varepsilon) > 0$ such that whenever $|y_0| < \gamma(\varepsilon)$ holds, it implies $|y(t)| < \varepsilon \quad \forall t \in [0, T_L]$;
- for any $y_0 \in \Omega \subseteq \mathbb{R}^n$ and $\{t_s\} \in \Lambda$, $\lim_{t \rightarrow T_L^-} |y(t)| = y(T_L) = 0$.

Meanwhile, if $\Omega = \mathbb{R}^n$, the system (1) is called uniformly globally P_S TS with respect to T_L over the class of impulsive sequence Λ .

Remark 2.1: The term 'uniformly' in Definition 2.1 is used to mean that the prescribed-time property of system (1) is independent of the choice of the impulsive sequence $\{t_s\} \in \Lambda$. Further, Definition 2.1 is different from several notions of stability like- fixed-time and predefined-time as the bound of the settling time can be preassigned arbitrarily and also does not depend on the parameters of the system respectively.

Definition 2.2 (X. Yang & Li, 2021): For a given impulsive time sequence $\{t_s\} \in \Lambda$, the function $V_L : [0, T_L] \times \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ is said to belong to the class ζ_0 if

- the function V_L is continuous on each of the sets $[t_{s-1}, t_s) \times \mathbb{R}^n$ and $\lim_{(t,z) \rightarrow (t_{s-1}^-, y)} V_L(t, y) = V_L(t_{s-1}^-, y)$ exists for all $y \in \mathbb{R}^n$ and $s - 1 \in \mathbb{N}$;
- V_L is Locally Lipschitz in y and $V_L(t, 0) = 0$ for all $t \in [0, T_L]$.

Definition 2.3 (Xing et al., 2024): Let $V_L \in \zeta_0$ and $(t, y) \in [t_{s-1}, t_s) \times \mathbb{R}^n$. Along the solution $y(t)$ of the system (1), the upper Dini derivative is described as

$$V'_{L+}(t, y(t)) = \limsup_{\varpi \rightarrow 0^+} \frac{V_L(t + \varpi, y(t) + \varpi f(t, y(t))) - V_L(t, y(t))}{\varpi}.$$

Definition 2.4 (He et al., 2022, κ_∞ Functions): A strictly increasing continuous function $\psi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is said to be a κ_∞ function if $\psi(0) = 0$ and $\psi(a) \rightarrow \infty$ as $a \rightarrow \infty$. And is denoted by $\psi \in \kappa_\infty$.

2.4 On an example

Example 2.1: Let's consider the following first-order time-varying impulse system:

$$\begin{cases} \dot{y}(t) = -\sqrt{y(t)} \\ (1 + y^2(t)) \frac{e^t}{e^2 - 1} + y(t) & t \geq 0, t \neq t_s, \\ (1 + y^2(t)) \frac{2e^t}{e^2 - 1}, \\ y(t_s) = \sqrt{\frac{P_1 y^2(t_s^-)}{1 + \frac{(1 - P_1)}{2} y^2(t_s^-)}}, & t = t_s, s \in \mathbb{N}. \end{cases} \quad (2)$$

Here, when one considers the case $P_1 = 1$, i.e. when there is no impulse in the system (2), the negative term of Equation (2) dominates over the positive term in some region around the origin and makes the system stable. Whereas, in some region away from the origin, the positive term dominates and makes the system unstable. So, it is fascinating to counter the effect of positive term present in Equation (2) through some proper designing of impulsive sequences and make the system (2) stable. In this case, we have considered $P_1 = \frac{1}{2}$ and can observe that the state of the system (2) reaches the origin in some finite time. Figure 1 validates the claims. Motivated by the aforementioned discussions, we are going to put forward some Lyapunov-based sufficient conditions for the prescribed-time stability of the system (1).

3. Main results

In this section, based on Lyapunov condition, two theorems are proposed for prescribed-time stability of system (1) consisting of stabilising and destabilising impulses, respectively.

Theorem 3.1: Let $T_L^\diamond = 2T_L > 0$ be a prescribed constant. If there exist functions $\psi_1, \psi_2 \in \kappa_\infty$, $V_L \in \zeta_0$; constants $l > 1$, $0 < p, p_1 < 1$, $\alpha > 0$ and continuous functions $\xi : [0, T_L^\diamond] \rightarrow \mathbb{R}_{\geq 0}$, $\mu : [0, T_L^\diamond] \rightarrow \mathbb{R}_+$ such that

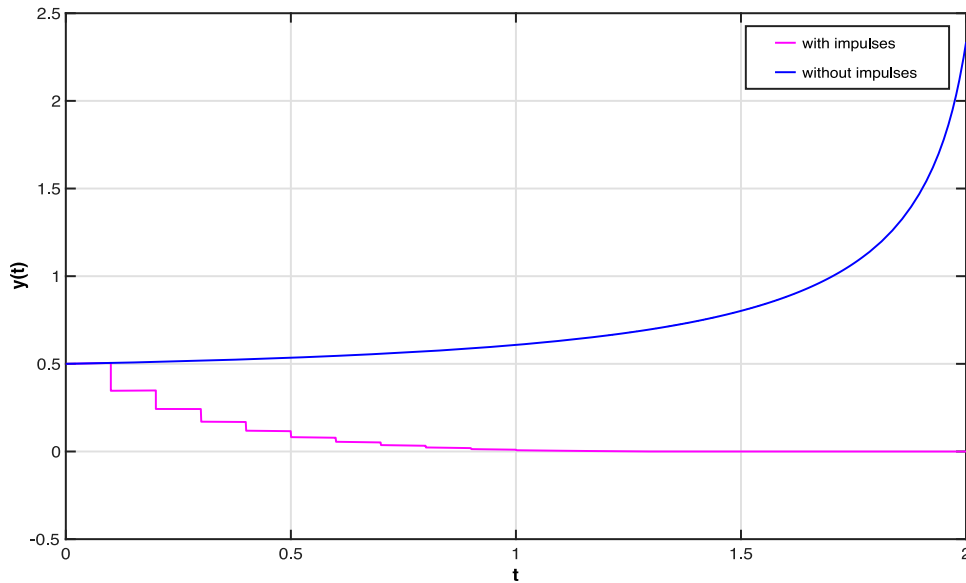


Figure 1. Simulation of Example 2.1.

- (i) $\psi_1(|y|) \leq V_L(t, y) \leq \psi_2(|y|)$, $\forall t \in [0, T_L^\diamond]$ and $\forall y \in \mathbb{R}^n$; for any $y_0 \in \mathbb{R}^n$, the solution $y(t) = y(t, 0, y_0)$ of system (1) satisfies
- (ii) $V_{L+}'(t, y(t)) \leq -\alpha^{p-1} V_L^p(t, y(t))(1 + \alpha V_L(t, y(t)))^{2-p} \mu(t) + V_L(t, y(t))(1 + \alpha V_L(t, y(t)))\xi(t)$, $t \neq t_s$;
- (iii) $V_L(t, g(t, y)) \leq \frac{p_1 V_L(t, y)}{1 + \alpha(1-p_1)V_L(t, y)}$, $\forall t \in [0, T_L^\diamond]$ and $y \in \mathbb{R}^n$;
- (iv) $\sup_{t \in [0, T_L^\diamond]} \frac{\xi(t)}{\mu(t)} = l$, then system (1) is uniformly globally P_S TS with respect to $T_L^\diamond$ over Λ_0 , if
- (v) $\int_0^{T_L} \mu(t) dt = 1$, and $\int_0^{T_L^\diamond} \mu(t) dt < \infty$, where $\Lambda_0 \subseteq \Lambda$ is the set of impulsive sequences $\{t_s\}$ with N as the minimum number of impulses satisfying
- (vi) $N = \min \left\{ i \in \mathbb{N} : i \geq \frac{\frac{1}{1-p} \ln(kl) - \int_0^{T_L} \xi(t) dt}{\ln(p_1)} \right\}$ with $k = \frac{1-p}{l(1+l-p)}$ and $t_N \leq T_L$.

Proof: For any given initial condition $y_0 \in \mathbb{R}^n$ and impulsive sequence $\{t_s\} \in \Lambda_0$, let $y(t) = y(t, 0, y_0)$ be the solution of system (1) through $(0, y_0)$. Further, let us consider $V_L(t) = V_L(t, y(t))$ and $V_{L0} = V_L(0, y_0)$. Here, the proof is divided into two parts as follows:

Case 1:

$$V_{L0} \leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}}.$$

The claim in this case is that

$$V_L(t) \leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}}$$

for all $t \in [0, T_L^\diamond]$. If this statement is not true, then there exist some $t \in [0, T_L^\diamond]$ such that

$$V_L(t) > \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}}.$$

Now, let

$$t^\diamond = \inf \left\{ t \in [0, T_L^\diamond] : V_L(t) > \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}} \right\}.$$

Also, from condition (iii) one can conclude that $t^\diamond$ is not an impulsive point. Hence, we have

$$V_L(t^\diamond) = \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}}, \quad V_L(t) \leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}}$$

for all $t \in [0, t^\diamond]$ and $V_{L+}'(t) |_{t=t^\diamond} \geq 0$. But, from conditions (ii) and (iv), one can have

$$\begin{aligned} V_{L+}'(t) |_{t=t^\diamond} &\leq -\alpha^{p-1} V_L^p(t^\diamond) (1 + \alpha V_L(t^\diamond))^{2-p} \mu(t^\diamond) \\ &\quad + V_L(t^\diamond) (1 + \alpha V_L(t^\diamond)) \xi(t^\diamond) \\ &= \alpha^{p-1} V_L^p(t^\diamond) (1 + \alpha V_L(t^\diamond))^{2-p} \mu(t^\diamond) \\ &\quad \times \left\{ -1 + \alpha^{1-p} V_L^{1-p}(t^\diamond) (1 + \alpha V_L(t^\diamond))^{p-1} \frac{\xi(t^\diamond)}{\mu(t^\diamond)} \right\} \\ &\leq \alpha^{p-1} V_L^p(t^\diamond) (1 + \alpha V_L(t^\diamond))^{2-p} \mu(t^\diamond) \{-1 + kl^2\} \\ &= -\frac{1}{(1+l-lp)} \alpha^{p-1} V_L^p(t^\diamond) (1 + \alpha V_L(t^\diamond))^{2-p} \mu(t^\diamond) \\ &< 0. \end{aligned}$$

Thus we arrive at a contradiction and hence the claim

$$V_L(t) \leq V_{L0} = \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}}$$

for all $t \in [0, T_L^\diamond]$ stands true. Now for any $\epsilon > 0$, $y_0 \in \mathbb{R}^n$ and $\{t_s\} \in \Lambda_0$ we can choose some $\delta \leq \psi_2^{-1}(\psi_1(\epsilon))$ such that whenever $|y_0| \leq \delta$ it implies $|y(t)| \leq \epsilon$. So, system (1) is uniformly

globally Lyapunov stable. Now, we are heading to show that it is not only Lyapunov stable but P_3TS in prescribed time $T_L^\diamond$.

For this, we have for all $t \in [0, T_L^\diamond]$

$$\begin{aligned}
 V_{L+}'(t) &\leq -\alpha^{p-1} V_L^p(t) (1 + \alpha V_L(t))^{2-p} \mu(t) \\
 &\quad + V_L(t) (1 + \alpha V_L(t)) \xi(t) \\
 &\leq \alpha^{p-1} V_L^p(t) (1 + \alpha V_L(t))^{2-p} \mu(t) \\
 &\quad \times \left\{ -1 + \alpha^{1-p} V_L^{1-p}(t) (1 + \alpha V_L(t))^{p-1} \right. \\
 &\quad \left. \times \sup_{[0, T_L^\diamond]} \frac{\xi(t)}{\mu(t)} \right\} \\
 &= \alpha^{p-1} V_L^p(t) (1 + \alpha V_L(t))^{2-p} \mu(t) \{-1 + kl^2\} \\
 &= -\frac{1}{(1+l-lp)} \alpha^{p-1} V_L^p(t) (1 + \alpha V_L(t))^{2-p} \mu(t).
 \end{aligned} \tag{3}$$

From Equation (3), one can obtain

$$\begin{aligned}
 \int_{V_{L0}}^{V_L(t)} \frac{\alpha^{p-1} U^{-p}}{(1 + \alpha U)^{2-p}} dU &\leq -\frac{1}{(1+l-lp)} \int_0^t \mu(w) dw \\
 \text{i.e. } \int_{V_{L0}}^{V_L(t)} \left(\frac{\alpha U}{1 + \alpha U} \right)^{-p} d \left(\frac{\alpha U}{1 + \alpha U} \right) \\
 &\leq -\frac{1}{(1+l-lp)} \int_0^t \mu(w) dw \\
 \text{i.e. } \left\{ 1 - \frac{1}{1 + \alpha V_L(t)} \right\}^{1-p} &\leq \left\{ 1 - \frac{1}{1 + \alpha V_{L0}} \right\}^{1-p} \\
 &\quad - \frac{(1-p)}{(1+l-lp)} \int_0^t \mu(w) dw \\
 \text{i.e. } \left\{ 1 - \frac{1}{1 + \alpha V_L(t)} \right\}^{1-p} &\leq \frac{(1-p)}{(1+l-lp)} \\
 &\quad - \frac{(1-p)}{(1+l-lp)} \int_0^t \mu(w) dw.
 \end{aligned} \tag{4}$$

Since, $\int_0^{T_L} \mu(t) dt = 1$, we get

$$\begin{aligned}
 \lim_{t \rightarrow T_L^-} \psi_1(|y(t)|) &\leq \lim_{t \rightarrow T_L^-} V_L(t) = V_L(T_L) = 0 \\
 \text{i.e. } \lim_{t \rightarrow T_L^-} |y(t)| &= y(T_L) = 0.
 \end{aligned} \tag{5}$$

So, the trajectories of system (1) reach the origin within the time $T_L < T_L^\diamond$ (prescribed time).

Case 2:

$$V_{L0} > \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}}.$$

From condition (ii), one can obtain

$$V_{L+}'(t) \leq V_L(t) (1 + \alpha V_L(t)) \xi(t), \quad \forall t \in [0, T_L]. \tag{6}$$

Which on simplification gives

$$\frac{\alpha V_L(t)}{1 + \alpha V_L(t)} \leq \frac{\alpha V_{L0}}{1 + \alpha V_{L0}} e^{\int_0^t \xi(w) dw}, \quad \forall t \in [0, t_1].$$

Moreover, from condition (iii), one can have at $t = t_1$

$$\begin{aligned}
 V_L(t_1) &\leq \frac{p_1 V_L(t_1^-)}{1 + (1-p_1) \alpha V_L(t_1^-)} \\
 \text{i.e. } \frac{\alpha V_L(t_1)}{1 + \alpha V_L(t_1)} &\leq p_1 \frac{\alpha V_{L0}}{1 + \alpha V_{L0}} e^{\int_0^{t_1} \xi(w) dw}.
 \end{aligned} \tag{7}$$

Combining Equation (6) with Equation (7), one can obtain

$$\frac{\alpha V_L(t)}{1 + \alpha V_L(t)} \leq p_1 \frac{\alpha V_{L0}}{1 + \alpha V_{L0}} e^{\int_0^t \xi(w) dw}, \quad \forall t \in [t_1, t_2].$$

Continuing in similar manner one can acquire

$$\frac{\alpha V_L(t)}{1 + \alpha V_L(t)} \leq p_1^{N-1} \frac{\alpha V_{L0}}{1 + \alpha V_{L0}} e^{\int_0^t \xi(w) dw}, \quad \forall t \in [t_{N-1}, t_N].$$

Now, at $t = t_N$, we have

$$\begin{aligned}
 V_L(t_N) &\leq \frac{p_1 V_L(t_N^-)}{1 + (1-p_1) \alpha V_L(t_N^-)} \\
 \text{i.e. } \frac{\alpha V_L(t_N)}{1 + \alpha V_L(t_N)} &\leq p_1^N \frac{\alpha V_{L0}}{1 + \alpha V_{L0}} e^{\int_0^{t_N} \xi(w) dw} \\
 &\leq p_1^N e^{\int_0^{t_N} \xi(w) dw}.
 \end{aligned} \tag{8}$$

Further, from definition of N , one can have

$$\frac{\alpha V_L(t_N)}{1 + \alpha V_L(t_N)} \leq (kl)^{\frac{1}{(1-p)}} e^{-\int_0^{T_L} \xi(w) dw} e^{\int_0^{t_N} \xi(w) dw}. \tag{9}$$

Meanwhile using the condition $t_N \leq T_L$, Equation (9) reduces to

$$\begin{aligned}
 \frac{\alpha V_L(t_N)}{1 + \alpha V_L(t_N)} &\leq (kl)^{\frac{1}{(1-p)}} \\
 \text{i.e. } V_L(t_N) &\leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{(1-p)}}}{1 - (kl)^{\frac{1}{(1-p)}}}.
 \end{aligned} \tag{10}$$

Thus, in this case system (1) reaches the boundary of the stable region, described in **Case 1** within some time $t_N \leq T_L$. Afterwards, similar procedure as **Case 1** can be done to make the system (1) reach the origin within some time T_L . Hence, total maximum time to reach the trajectories of system (1) is $T_L + T_L = T_L^\diamond$. So, the system (1) is uniformly globally P_3TS with respect to $T_L^\diamond$ over Λ_0 . This completes the proof of Theorem 3.1. ■

Remark 3.1: From condition (ii) of Theorem 3.1, one can recognise that the presence of the positive term $V_L(t, y(t))(1 + \alpha V_L(t, y(t))) \xi(t)$ in the Lyapunov inequality makes it indefinite. Thus there exists some regions where the term $(V_L(t, y(t))(1 + \alpha V_L(t, y(t))) \xi(t))$ in the Lyapunov inequality dominates over the negative one $(-\alpha^{p-1} V_L^p(t, y(t))(1 + \alpha V_L(t, y(t)))^{2-p} \mu(t))$ and makes the system unstable (refer to Example 2.1 and Section 5). Hence, this type of Lyapunov inequality increases its

domain of applicability and is more practical in control engineering. Considering all of these into account, Theorem 3.1 provides a sufficient condition for uniformly and globally P_S TS of nonlinear time-varying impulsive systems by the use of stabilising impulses and time-varying gain functions. Elaborately, the conditions on the time-varying gain function $\mu(t)$ are essential to obtain negative definiteness of the constructed Lyapunov inequality within the domain

$$\gamma_1 = \left\{ y \in \mathbb{R}^n : V_L(t, y) \leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}} \right\},$$

and further to achieve $y(T_L) = 0$ (refer to Equations 4 and 5). Whereas the conditions on the function $\xi(t)$ are used to obtain the minimum number of impulses N , corresponding to **Case 2** (refer to Equations 8 and 9), along with acquiring negative definiteness of the constructed Lyapunov inequality within the domain

$$\gamma_1 = \left\{ y \in \mathbb{R}^n : V_L(t, y) \leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}} \right\}.$$

Moreover, the whole domain $\mathbb{R}^n$ is divided into two parts:

$$\gamma_1 = \left\{ y \in \mathbb{R}^n : V_L(t, y) \leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}} \right\}$$

and

$$\gamma_2 = \left\{ y \in \mathbb{R}^n : V_L(t, y) > \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}} \right\}.$$

In γ_1 , the state of the system (1) is uniformly P_S TS within the time T_L , and there is no impulse in this case. But, when the states are in the region γ_2 , the trajectories of system (1) may be unstable due to the presence of the term ' $V_L(t, y(t))(1 + \alpha V_L(t, y(t)))\xi(t)$ '. In this case, we have constructed the stabilising impulses, satisfying certain conditions to bring the trajectories into the region γ_1 within the time T_L .

Remark 3.2: From the condition (4) of Theorem 1 of Xing et al. (2024), one can observe that the position of the impulsive points is also required along with the minimum number of impulses for F TS of the corresponding system. But, here such restrictions on the positions of the impulses are not required for P_S TS.

Remark 3.3: He et al. (2022) have given a result on local P_S TS for autonomous system via impulsive regulation. In contrast, in Theorem 3.1 of the present article, we have provided uniform global P_S TS result for non-autonomous systems via properly designing impulsive sequences and considering time-varying gain function.

Remark 3.4: Most of the articles on P_S TS have considered the interval $[0, T_L^\diamond)$ and the integration of the gain function $\mu(t)$ over the interval $[0, T_L^\diamond)$ used to be infinite. Thus, as mentioned

by Xu and Liu (2023), it is required to establish the validity of $\dot{V}_L(T_L^\diamond) = 0$. This part is missing in almost all of the available literature. To eliminate this problem, we have considered the interval $[0, T_L^\diamond]$ along with the gain function $\mu(t)$ giving finite value on integration over the interval $[0, T_L^\diamond]$.

Theorem 3.2: Let $T_L > 0$ be a prescribed constant. If there exist functions $\psi_1, \psi_2 \in \kappa_\infty$, $V_L \in \zeta_0$; constants $l > 1, \sigma \in (0, 1)$, $0 < p < 1, p_1 > 1, \alpha > 0$ and continuous functions $\xi : [0, T_L] \rightarrow \mathbb{R}_{\geq 0}$, $\mu : [0, T_L] \rightarrow \mathbb{R}_+$ such that

- (i) $\psi_1(|y|) \leq V_L(t, y) \leq \psi_2(|y|)$, $\forall t \in [0, T_L]$ and $\forall y \in \mathbb{R}^n$; for any $y_0 \in \mathbb{R}^n$, the solution $y(t) = y(t, 0, y_0)$ of system (1) satisfies
- (ii) $V'_{L+}(t, y(t)) \leq -\alpha^{p-1} V_L(t, y(t))(1 + \alpha V_L(t, y(t)))^{2-p} \mu(t) + V_L(t, y(t))(1 + \alpha V_L(t, y(t)))\xi(t)$, $t \neq t_s$;
- (iii) $V_L(t, g(t, y)) \leq p_1 V_L(t, y)$, $\forall t \in [0, T_L]$ and $y \in \mathbb{R}^n$;
- (iv) $\sup_{t \in [0, T_L]} \frac{\xi(t)}{\mu(t)} = l$.

Then system (1) is uniformly locally P_S TS with respect to T_L in

$$\Delta = \left\{ y \in \mathbb{R}^n : |y| \leq \psi_2^{-1} \left(\frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{p_1 - (kl)^{\frac{1}{1-p}}} \right) \right\}$$

with $k = \frac{1-p}{l(1+l-p)}$ over $\Lambda_\oplus$ if $\int_0^{T_L} \mu(t) dt = 1$, where $\Lambda_\oplus \subseteq \Lambda$ is the set of impulsive classes $\{t_s\}$ with N_0 as the maximum number of impulses satisfying

- (a) $\int_0^{t_1} \mu(w) dw \geq \frac{1-\sigma}{p_1}$;
- (b) $\int_{t_{i-1}}^{t_i} \mu(w) dw \geq \frac{\sigma(p_1^{1-p}-1)}{p_1}$, for $i \in \{2, 3, \dots, N_0\}$;
- (c) $\int_{t_{N_0}}^{T_L} \mu(w) dw \geq \sigma$;

with $t_{N_0} < T_L$ and

- (i1) $N_0 \leq \frac{1}{\sigma}$, if all the terms of conditions (a), (b) and (c) are equality.
- (i2) $N_0 < \frac{1}{\sigma}$, if at least one of the terms of conditions (a), (b) and (c) is strict inequality.

Proof: For any given initial condition $y_0 \in \Delta$ and impulsive sequence $\{t_s\} \in \Lambda_\oplus$, let $y(t) = y(t, 0, y_0)$ be the solution of system (1) through $(0, y_0)$. Further, without any loss of generality let us consider $V_L(t) = V_L(t, y(t))$ and $V_{L0} = V_L(0, y_0)$. Moreover, $y_0 \in \Delta$ and condition (i) together gives

$$V_{L0} \leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{p_1 - (kl)^{\frac{1}{1-p}}}.$$

The proof of this theorem is divided into two parts, depending on $N_0 = 0$ or $N_0 \geq 1$, as follows:

Case I: $N_0 = 0$ i.e. there is no destabilising impulses.

In this case, we have

$$V_{L0} \leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{p_1 - (kl)^{\frac{1}{1-p}}} \leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{1-p}}}{1 - (kl)^{\frac{1}{1-p}}}. \quad (11)$$

Thus, combining Equation (11) with condition (ii) and using the same procedure as **Case 1** of Theorem 3.1 one can obtain $\lim_{t \rightarrow T_L^-} |y(t)| = y(T_L) = 0$. Also, for any $\epsilon > 0$, $y_0 \in \Delta$ and $\{t_s\} \in \Lambda_\oplus$ we can choose some $\delta \leq \psi_2^{-1}(\psi_1(\epsilon))$ such that whenever $|y_0| \leq \delta$ it implies $|y(t)| \leq \epsilon$.

Case II: $N_0 \geq 1$, i.e. there is at least one destabilising impulse. It can be seen easily from **Case 1** of Theorem 3.1 that

$$V_L(t) \leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{(1-p)}}}{1 - (kl)^{\frac{1}{(1-p)}}}$$

for all $t \in [0, t_1)$. Hence, from conditions (ii) and (iv), we have

$$V'_{L+}(t) \leq -\frac{1}{(1+l-pl)} \alpha^{p-1} V_L^p(t) (1 + \alpha V_L(t))^{2-p} \mu(t),$$

$$\forall t \in [0, t_1),$$

which on integration gives

$$\left(\frac{\alpha V_L(t)}{1 + \alpha V_L(t)} \right)^{1-p} \leq \left(\frac{\alpha V_{L0}}{1 + \alpha V_{L0}} \right)^{1-p}$$

$$- \frac{(1-p)}{(1+l-pl)} \int_0^t \mu(w) dw,$$

$$t \in [0, t_1). \quad (12)$$

Also from condition (iii), at the impulsive point $t = t_1$ we have

$$\left(\frac{\alpha V_L(t_1)}{1 + \alpha V_L(t_1)} \right)^{1-p} \leq p_1^{1-p} \left(\frac{\alpha V_L(t_1^-)}{1 + \alpha V_L(t_1^-)} \right)^{1-p}. \quad (13)$$

Combining Equations (12) and (13), one can have

$$\left(\frac{\alpha V_L(t_1)}{1 + \alpha V_L(t_1)} \right)^{1-p} \leq p_1^{1-p} \left(\frac{\alpha V_{L0}}{1 + \alpha V_{L0}} \right)^{1-p}$$

$$- p_1^{1-p} \frac{(1-p)}{(1+l-pl)} \int_0^{t_1} \mu(w) dw$$

$$\leq kl - p_1^{1-p} \frac{(1-p)}{(1+l-pl)} \int_0^{t_1} \mu(w) dw$$

$$= kl - p_1^{1-p} kl \int_0^{t_1} \mu(w) dw. \quad (14)$$

Using condition (a) in Equation (14), we get

$$\left(\frac{\alpha V_L(t_1)}{1 + \alpha V_L(t_1)} \right)^{1-p} \leq \sigma kl < kl \quad (15)$$

$$\text{i.e. } V_L(t_1) \leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{(1-p)}}}{1 - (kl)^{\frac{1}{(1-p)}}}. \quad (16)$$

Proceeding similarly for the case of $t \in [0, t_1)$, we get

$$\left(\frac{\alpha V_L(t)}{1 + \alpha V_L(t)} \right)^{1-p} \leq \left(\frac{\alpha V_L(t_1)}{1 + \alpha V_L(t_1)} \right)^{1-p}$$

$$- kl \int_{t_1}^t \mu(w) dw, \quad t \in [t_1, t_2)$$

$$\leq \sigma kl - kl \int_{t_1}^t \mu(w) dw, \quad t \in [t_1, t_2). \quad (17)$$

Also, from condition (iii), at the impulsive point $t = t_2$ one has

$$\left(\frac{\alpha V_L(t_2)}{1 + \alpha V_L(t_2)} \right)^{1-p} \leq p_1^{1-p} \left(\frac{\alpha V_L(t_2^-)}{1 + \alpha V_L(t_2^-)} \right)^{1-p}. \quad (18)$$

Hence, using Equation (17) in Equation (18) it is easy to obtain that

$$\left(\frac{\alpha V_L(t_2)}{1 + \alpha V_L(t_2)} \right)^{1-p} \leq p_1^{1-p} \sigma kl - p_1^{1-p} kl \int_{t_1}^{t_2} \mu(w) dw. \quad (19)$$

From Equation (19) along with the condition (b), it can be easily obtained that

$$\left(\frac{\alpha V_L(t_2)}{1 + \alpha V_L(t_2)} \right)^{1-p} \leq \sigma kl < kl$$

$$\text{i.e. } V_L(t_2) \leq \frac{1}{\alpha} \frac{(kl)^{\frac{1}{(1-p)}}}{1 - (kl)^{\frac{1}{(1-p)}}}. \quad (20)$$

Proceeding in similar manner as above, one can have for $t \in [t_{N_0-1}, t_{N_0})$

$$\left(\frac{\alpha V_L(t)}{1 + \alpha V_L(t)} \right)^{1-p} \leq \left(\frac{\alpha V_L(t_{N_0-1})}{1 + \alpha V_L(t_{N_0-1})} \right)^{1-p}$$

$$- kl \int_{t_{N_0-1}}^t \mu(w) dw,$$

and at the impulsive point $t = t_{N_0}$

$$\left(\frac{\alpha V_L(t_{N_0})}{1 + \alpha V_L(t_{N_0})} \right)^{1-p} \leq \sigma kl. \quad (21)$$

Now, consider the interval $[t_{N_0}, T_L]$. Here also we can easily have

$$\left(\frac{\alpha V_L(t)}{1 + \alpha V_L(t)} \right)^{1-p} \leq \left(\frac{\alpha V_L(t_{N_0})}{1 + \alpha V_L(t_{N_0})} \right)^{1-p}$$

$$- kl \int_{t_{N_0}}^t \mu(w) dw. \quad (22)$$

On using Equations (21), (22) reduces to

$$\left(\frac{\alpha V_L(t)}{1 + \alpha V_L(t)} \right)^{1-p} \leq \sigma kl - kl \int_{t_{N_0}}^t \mu(w) dw. \quad (23)$$

By the use of condition (c) in Equation (23), one can easily conclude that $\lim_{t \rightarrow T_L^-} V_L(t) = V_L(T_L) = 0$ as $t \rightarrow T_L^-$.

One point here to be noted that, since $\int_0^{T_L} \mu(t) dt = 1$, one should have

$$\int_0^{t_1} \mu(w) dw + \sum_{i=2}^{N_0} \left\{ \int_{t_{i-1}}^{t_i} \mu(w) dw \right\} + \int_{t_{N_0}}^{T_L} \mu(w) dw = 1. \quad (24)$$

Combining conditions (a),(b) and (c), Equation (24) easily gives

- (i1) $N_0 \leq \frac{1}{\sigma}$, if all the terms of conditions (a),(b) and (c) are equality.
- (i2) $N_0 < \frac{1}{\sigma}$, if at least one of the terms of conditions (a),(b) and (c) is strict inequality.

Thus, for both the cases it has been shown that the system (1) is uniformly locally P_S TS with respect to T_L and $\Delta = \left\{ y \in \mathbb{R}^n : |y| \leq \psi_2^{-1} \left(\frac{1}{\alpha} \frac{(kl)^{\frac{1}{(1-p)}}}{p_1 - (kl)^{\frac{1}{(1-p)}}} \right) \right\}$ with $k = \frac{1-p}{l(1+l-p)}$ over $\Lambda_{\oplus}$. This completes the proof of the Theorem 3.2. ■

Remark 3.5: Theorem 3.2 provides a sufficient condition for uniform local P_S TS of system (1) with destabilising impulses. In detail, we have constructed a region Δ which is P_S TS. If one considers initial conditions outside the region Δ , then the trajectories of the system (1) may be unstable due to the presence of the term ' $V_L(t, y(t))(1 + \alpha V_L(t, y(t)))\xi(t)$ '. Hence, providing destabilising impulses in this region will increase the possibility of instability of the trajectories. So, we confined ourselves to the region Δ and tried to find the maximum number of destabilising impulses that can be applied without effecting its P_S TS properties. From conditions (i1) and (i2), it can be observed that for a given value of σ , the maximum number of destabilising impulses cannot exceed $\frac{1}{\sigma}$. Moreover, as $\sigma \in (0, 1)$, there is scope of at least one destabilising impulse. Also, decreasing the value of σ increases the scope of more number of destabilising impulses and vice versa. Further, $p, P_1, \mu(t)$ are also known. Now, if one can obtain a t_1 such that the condition (a) holds, then only there can be one destabilising impulse, and similarly for conditions (b) and (c). So the process of finding maximum number of destabilising impulses and their positions can be summarised as follows:

- Depending on the value of σ , one will have an idea of maximum number of destabilising impulses that can be there within the time interval $(0, T_L)$. It should be noted that the restriction on number of destabilising impulses given only by conditions (i1) and (i2) does not guarantee the existence of N_0 as the maximum number of possible destabilising impulses. For example, $\sigma = 0.9$ imply $N_0 \leq 1$ (from conditions (i1) and (i2)). But, if one cannot find any t_1 such that the condition (a) holds then it suffice $N_0 = 0$ as the maximum number of impulses, and not $N_0 = 1$, possible within the given interval $(0, T_L)$. So, not only the conditions (i1) and (i2), but the conditions (a), (b), and (c), along with it, provide the maximum number of destabilising impulses.

Remark 3.6: It should be noted that, in contrast to Theorem 3.1, where the position of the impulses was not needed to be specified (except $t_N \leq T_L$), Theorem 3.2 requires the position of the impulses (see conditions (a),(b) and (c)).

Remark 3.7: In contrast to the several recent articles on prescribed-time control (H. Wang, Zong, Yang, et al., 2024; Y. Wang, Zong, Zhao, et al., 2024; Zong et al., 2024), which are based on continuous control and can reach to some neighbourhood of the origin within some prescribed-time or can reach to

the origin asymptotically, the present study is dedicated to establish several results on prescribed-time stability (i.e. can reach exactly to the origin within the prescribed time) of impulsive systems constituting time-varying gain function. Here, the primary emphasis has been on improving the stability results on hybrid systems (continuous and discrete) to prescribed time. Moreover, the construction of Lyapunov-based sufficient conditions along with the formation of impulsive sequences to make the impulsive system stable within the prescribed time, in presence of stabilising and destabilising impulses were the main challenges of the work.

4. Application

This section is devoted to the application of stability results obtained in Section 3 to the prescribed-time stability of a class of time-varying affine system.

Let us examine the subsequent time-varying impulsive system

$$\begin{cases} \dot{y}(t) = f_a(t, y(t)) + g_a(t, y(t)) \hat{u}_a(t), & t \geq 0, t \neq t_s, \\ y(t) = f_b(t^-, y(t^-)) + g_b(t^-, y(t^-)) u_b(t^-), & t = t_s, \\ y(0) = y_0, \end{cases} \quad (25)$$

where $y \in \mathbb{R}^n$ is the state vector with initial condition y_0 ; f_a, g_a, f_b, g_b are continuous functions from $\mathbb{R}_{\geq 0} \times \mathbb{R}^n$ to $\mathbb{R}^n, \mathbb{R}^{n \times m_a}, \mathbb{R}^n, \mathbb{R}^{n \times m_b}$ respectively. Further, $f_a(t, 0) = f_b(t, 0) = 0, \forall t \geq 0$; $\hat{u}_a \in \mathbb{R}^{m_a}$ and $u_b \in \mathbb{R}^{m_b}$ are the control inputs with $\hat{u}_a = u_a + \delta(t, y)$, where $\delta(t, y)$ is the uncertain disturbances.

Theorem 4.1: Consider the affine system (25) and let $T_L^\diamond = 2T_L > 0$ be a prescribed constant. If $g_a(t, y)$ and $g_b(t, y)$ are invertible functions and there exist constants $\alpha > 0, p, p_1 \in (0, 1)$, diagonal matrix-valued function $Q(t) : [0, T_L^\diamond] \rightarrow \mathbb{R}^{n \times n}$, continuous functions $\xi : [0, T_L^\diamond] \rightarrow \mathbb{R}_{\geq 0}, \mu : [0, T_L^\diamond] \rightarrow \mathbb{R}_{+}$ such that

- (i) $\tau(t, y)\delta(t, y) \leq \{[\text{sgn}(y)]^T Q(t)y\}\xi(t)$.
- (ii) $\sup_{t \in [0, T_L^\diamond]} \frac{\xi(t)}{\mu(t)} = l$.
- (iii) $\int_0^{T_L} \mu(t) dt = 1$.

Then the given system (25) is uniformly globally P_S TS with respect to $T_L^\diamond$ over Λ_0 under the feedback control $(u_a, u_b) = (\Gamma_a(t, y), \Gamma_b(t, y))$ given by

$$\begin{aligned} \Gamma_a(t, y) &= -g_a^{-1} Q^{-1}(t, y) [\text{sgn}(y)] \\ &\quad \times \left\{ \chi(t, y) + [\text{sgn}(y)]^T \dot{Q}(t)y \right. \\ &\quad \left. + \alpha^{p-1} \left\{ [\text{sgn}(y)]^T Q(t)y \right\}^p \right. \\ &\quad \left. \times \left\{ 1 + \alpha [\text{sgn}(y)]^T Q(t)y \right\}^{2-p} \mu(t) \right\}, \end{aligned} \quad (26)$$

and

$$\Gamma_b(t, y) = g_b^{-1} \left\{ -f_b + \frac{p_1 y}{1 + (\alpha - p_1 \alpha) [\text{sgn}(y)]^T Q(t)y} \right\}, \quad (27)$$

where $\tau(t, y) = [\text{sgn}(y)]^T Q(t) g_a$, $\chi(t, y) = [\text{sgn}(y)]^T Q(t) f_a$ and $\Lambda_0 \subseteq \Lambda$ is the set of impulsive sequences $\{t_s\}$ satisfying $N = \min \left\{ i \in \mathbb{N} : i \geq \frac{\frac{1}{1-p} \ln(kl) - \int_0^{T_L} \xi(t) dt}{\ln(p_1)} \right\}$ with $k = \frac{1-p}{l(1+l-p)}$ and $t_N \leq T_L$.

Proof: Given $T_L > 0$, for any sequence $\{t_s\} \in \Lambda_0$ and initial condition $y_0 \in \mathbb{R}^n$, let $y(t) = y(t, 0, y_0)$ be the solution of system (25) through $(0, y_0)$. Let $V_L(t) = V_L(t, y(t)) = [\text{sgn}(y(t))]^T Q(t) y(t)$ be the Lyapunov function. Then, upper right-hand Dini derivative of $V_L(t)$ along the solution $y(t)$ of system (25) at $t \neq t_s$ satisfies

$$\begin{aligned} V'_{L+}(t) &= [\text{sgn}(y(t))]^T Q(t) \dot{y}(t) + [\text{sgn}(y(t))]^T \dot{Q}(t) y(t) \\ &= [\text{sgn}(y(t))]^T Q(t) \{f_a(t, y(t)) \\ &\quad + g_a(t, y(t)) (u_a + \delta(t, y))\} \\ &\quad + [\text{sgn}(y(t))]^T \dot{Q}(t) y(t) \\ &\leq \chi(t, y) + \tau(t, y) \Gamma_a(t, y) \\ &\quad + \left\{ [\text{sgn}(y(t))]^T Q(t) y(t) \right\} \xi(t) \\ &\quad + [\text{sgn}(y(t))]^T \dot{Q}(t) y(t). \end{aligned} \quad (28)$$

Using the feedback controller $\Gamma_a(t, y)$ from Equation (26) in Equation (28), one can obtain

$$\begin{aligned} V'_{L+}(t) &\leq -\alpha^{p-1} \left\{ [\text{sgn}(y)]^T Q(t) y \right\}^p \\ &\quad \times \left\{ 1 + \alpha [\text{sgn}(y)]^T Q(t) y \right\}^{2-p} \mu(t) \\ &\quad + \left\{ [\text{sgn}(y(t))]^T Q(t) y(t) \right\} \xi(t) \\ &= -\alpha^{p-1} V_L^p(t) \{1 + \alpha V_L(t)\}^{2-p} \mu(t) + V_L(t) \xi(t) \\ &\leq -\alpha^{p-1} V_L^p(t) \{1 + \alpha V_L(t)\}^{2-p} \mu(t) \\ &\quad + V_L(t) \{1 + \alpha V_L(t)\} \xi(t). \end{aligned} \quad (29)$$

Also, at the impulsive points $t = t_s$, we have

$$\begin{aligned} V_L(t_s) &= [\text{sgn}(y(t_s))]^T Q(t_s) y(t_s) \\ &= [\text{sgn}(f_b(t_s^-, y(t_s^-)) + g_b(t_s^-, y(t_s^-)) \\ &\quad \times \Gamma_b(t_s^-, y(t_s^-)))]^T Q(t_s) (f_b(t_s^-, y(t_s^-)) \\ &\quad + g_b(t_s^-, y(t_s^-)) \Gamma_b(t_s^-, y(t_s^-))). \end{aligned} \quad (30)$$

Using the impulsive controller from Equations (27), (30) reduces to

$$\begin{aligned} V_L(t_s) &= \left[\text{sgn} \left(\frac{p_1 y(t_s^-)}{1 + (\alpha - p_1 \alpha) [\text{sgn}(y(t_s^-))]^T Q(t_s^-) y(t_s^-)} \right) \right]^T \\ &\quad \times Q(t_s) \left(\frac{p_1 y(t_s^-)}{1 + (\alpha - p_1 \alpha) [\text{sgn}(y(t_s^-))]^T Q(t_s^-) y(t_s^-)} \right) \end{aligned}$$

$$\begin{aligned} &= \frac{p_1}{1 + (\alpha - p_1 \alpha) [\text{sgn}(y(t_s^-))]^T Q(t_s^-) y(t_s^-)} \\ &\quad \times \left\{ [\text{sgn}(y(t_s^-))]^T Q(t_s) y(t_s^-) \right\} \\ &= \frac{p_1 V_L(t_s^-)}{1 + (\alpha - p_1 \alpha) V_L(t_s^-)}. \end{aligned} \quad (31)$$

So, by virtue of Theorem 3.1, Equations (29) and (31), we have that the system (25) is uniformly globally P_S TS with respect to prescribed time $T_L^\diamond$ over the class of impulsive sequence Λ_0 .

This completes the proof of Theorem 4.1. ■

5. Numerical simulations

Three examples are given in this section of the manuscript to support the conclusions drawn in earlier sections.

Example 5.1: Let us examine the subsequent nonlinear system that varies over time:

$$\begin{cases} \dot{y}_1(t) = -\frac{1}{2} \frac{e^t}{e^{T_L} - 1} |y_1|^{\frac{1}{2}} (1 + y_1^2 + y_2^2 + y_3^2)^{\frac{5}{4}} \text{sgn}(y_1) \\ \quad + \frac{e^t}{e^{T_L} - 1} y_1 (1 + y_1^2 + y_2^2 + y_3^2) + y_2 y_3 + y_1 y_3^2, \\ \dot{y}_2(t) = -\frac{1}{2} \frac{e^t}{e^{T_L} - 1} |y_2|^{\frac{1}{2}} (1 + y_1^2 + y_2^2 + y_3^2)^{\frac{5}{4}} \text{sgn}(y_2) \\ \quad + \frac{e^t}{e^{T_L} - 1} y_2 (1 + y_1^2 + y_2^2 + y_3^2) + y_1 y_3, \\ \dot{y}_3(t) = -\frac{1}{2} \frac{e^t}{e^{T_L} - 1} |y_3|^{\frac{1}{2}} (1 + y_1^2 + y_2^2 + y_3^2)^{\frac{5}{4}} \text{sgn}(y_3) \\ \quad + \frac{e^t}{e^{T_L} - 1} y_3 (1 + y_1^2 + y_2^2 + y_3^2) - 2y_1 y_2 - y_1^2 y_3, \end{cases} \quad (32)$$

with impulses

$$y_i(t) = \left\{ \frac{\frac{1}{2} y_i^2(t^-)}{1 + \frac{1}{2} (y_1^2(t^-) + y_2^2(t^-) + y_3^2(t^-))} \right\}^{\frac{1}{2}}, \quad i = 1, 2, 3; t = t_s, \quad (33)$$

where $y = (y_1, y_2, y_3)^T \in \mathbb{R}^3$, $\{t_s\} \in \Lambda$. Consider the Lyapunov function $V_L(y) = y_1^2 + y_2^2 + y_3^2$. Then one can easily find that it satisfies conditions (ii), (iii), (iv) and (v) of Theorem 3.1 with $\alpha = 1, p = \frac{3}{4}, p_1 = \frac{1}{2}, l = 2, \mu(t) = \frac{e^t}{e^{T_L} - 1}$ and $\xi(t) = \frac{2e^t}{e^{T_L} - 1}$. Hence, systems (32) and (33) are uniformly globally P_S TS with respect to prescribed time T_L over the class of impulsive sequence Λ_0 satisfying condition (vi) of Theorem 3.1. For numerical simulations, we have considered $y_0 = (-0.2, -0.5, 0.6)^T$, $T_L = 1, T_L^\diamond = 2$ and then from condition (vi) we have obtained $N \geq 12$ and $t_N \leq 1$. The trajectories of system (32) without the impulsive control (33) are given in Figure 2(a). And it can be concluded that they do not converge to the origin within the prescribed time $T_L^\diamond = 2$. Now, for the controlling prospective, let us consider $N = 12$ and the impulsive sequence as $\{0.05, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 0.96, 0.99\}$. Figure 2(b) verifies that systems (32) and (33) are P_S TS in prescribed time $T_L^\diamond = 2$ under the above conditions. Whereas

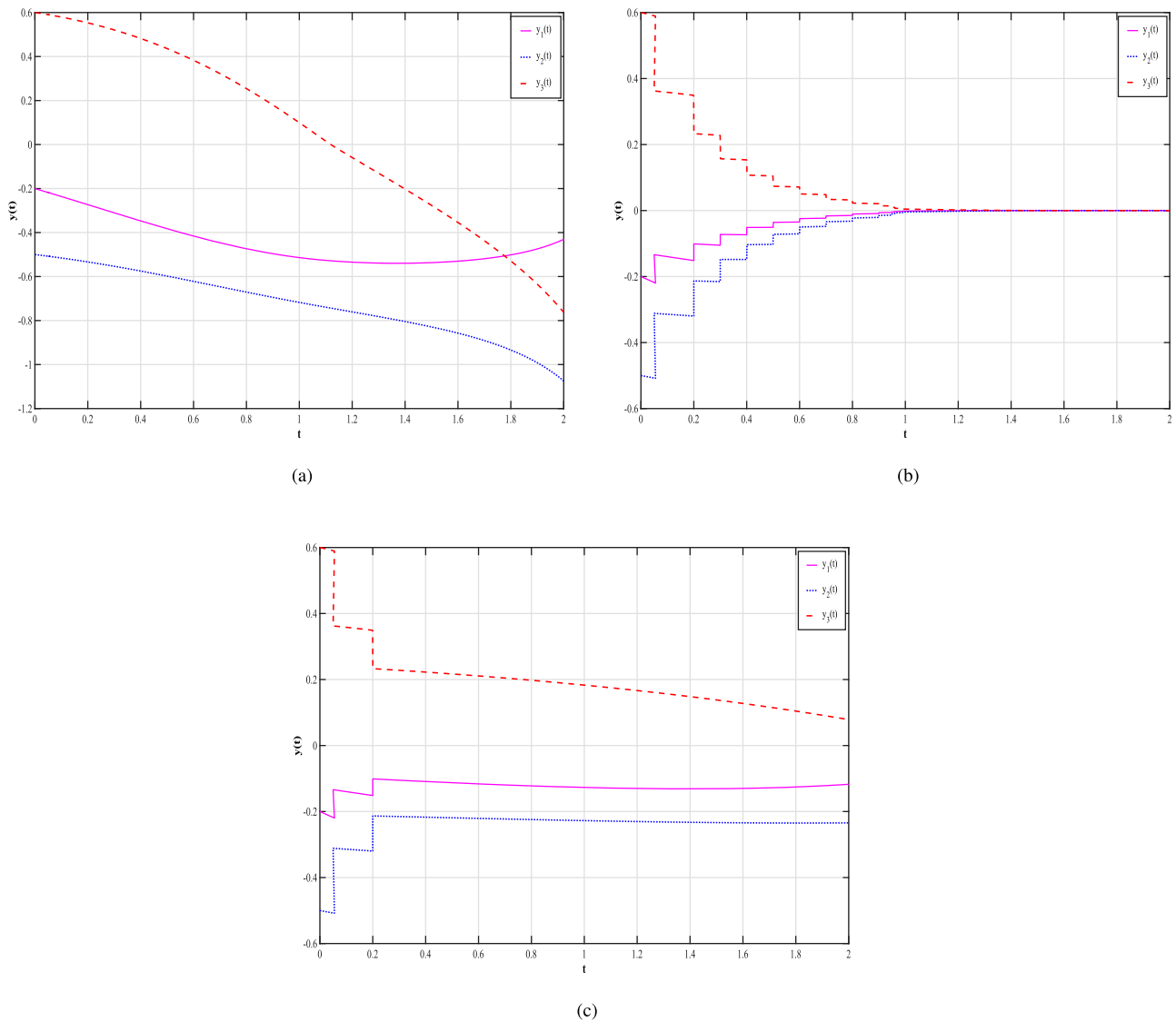


Figure 2. Simulations of Example 5.1.

Figure 2(c) shows that if we take $N = 2 < 12$ and impulsive points as $\{0.05, 0.2\}$, then under the same initial condition the above-mentioned system is not stable in prescribed time $T_L^\diamond = 2$. The above discussion verifies the statement of Theorem 3.1.

Example 5.2: Let us consider the subsequent time-varying nonlinear system

$$\begin{cases} \dot{y}_1(t) = y_1 |y_2| - \left(\frac{1}{2T_L} + \frac{t}{T_L^2} \right) |y_1|^{\frac{1}{2}} \times \\ \quad (1 + |y_1| + |y_2|)^{\frac{3}{2}} \operatorname{sgn}(y_1) \\ \quad + \left(\frac{1}{T_L} + \frac{2t}{T_L^2} \right) y_1 (1 + |y_1| + |y_2|), \\ \dot{y}_2(t) = -|y_1| |y_2| - \left(\frac{1}{2T_L} + \frac{t}{T_L^2} \right) |y_2|^{\frac{1}{2}} \times \\ \quad (1 + |y_1| + |y_2|)^{\frac{3}{2}} \operatorname{sgn}(y_2) \\ \quad + \left(\frac{1}{T_L} + \frac{2t}{T_L^2} \right) y_2 (1 + |y_1| + |y_2|), \end{cases} \quad (34)$$

with impulses

$$y_i(t) = p_1 y_i(t^-), i = 1, 2; t = t_s, \quad (35)$$

where $y = (y_1, y_2)^T \in \mathbb{R}^2$, $\{t_s\} \in \Lambda$. Consider $V_L(y) = |y_1| + |y_2|$ as the Lyapunov function, one can easily verify that it satisfies the conditions (i), (ii), (iii) and (iv) of Theorem 3.2 with $l = 2, p = \frac{1}{2}, p_1 = 2, \mu(t) = (\frac{1}{2T_L} + \frac{t}{T_L^2}), \xi(t) = (\frac{1}{T_L} + \frac{2t}{T_L^2}), \psi_1(r) = \psi_2(r) = |r|$. Thus systems (34) and (35) are uniformly locally $P_\delta TS$ with respect to prescribed time T_L over $\Delta = \{y \in \mathbb{R}^2 : |y_1| + |y_2| \leq \frac{1}{31}\}$ and $\Lambda_\oplus$ satisfying conditions (a), (b), (c), (i1), and (i2). Now, if we choose $\sigma = \frac{1}{2}$, then, by conditions (i1) and (i2) one have $N_0 \leq 2$. But, if we choose $t_1 = 0.5$, then condition (a) is a strict inequality. So, by condition (i2) we have $N_0 < 2$. For numerical simulation, we have considered $N_0 = 1, T_L = 1, y_0 = (0.03, -0.001)^T \in \Delta$. Figure 3(a) validates the truthfulness of the above statement. Moreover, if we consider $y_0 = (0.03, -0.1)^T \notin \Delta$, then Figure 3(b) validates that it is not stabilised in prescribed time $T_L = 1$. Further, if we consider $N_0 = 2$ and the impulsive sequence $\{0.01, 0.015\}$, i.e. it does not belong to $\Lambda_\oplus$, then one can observe

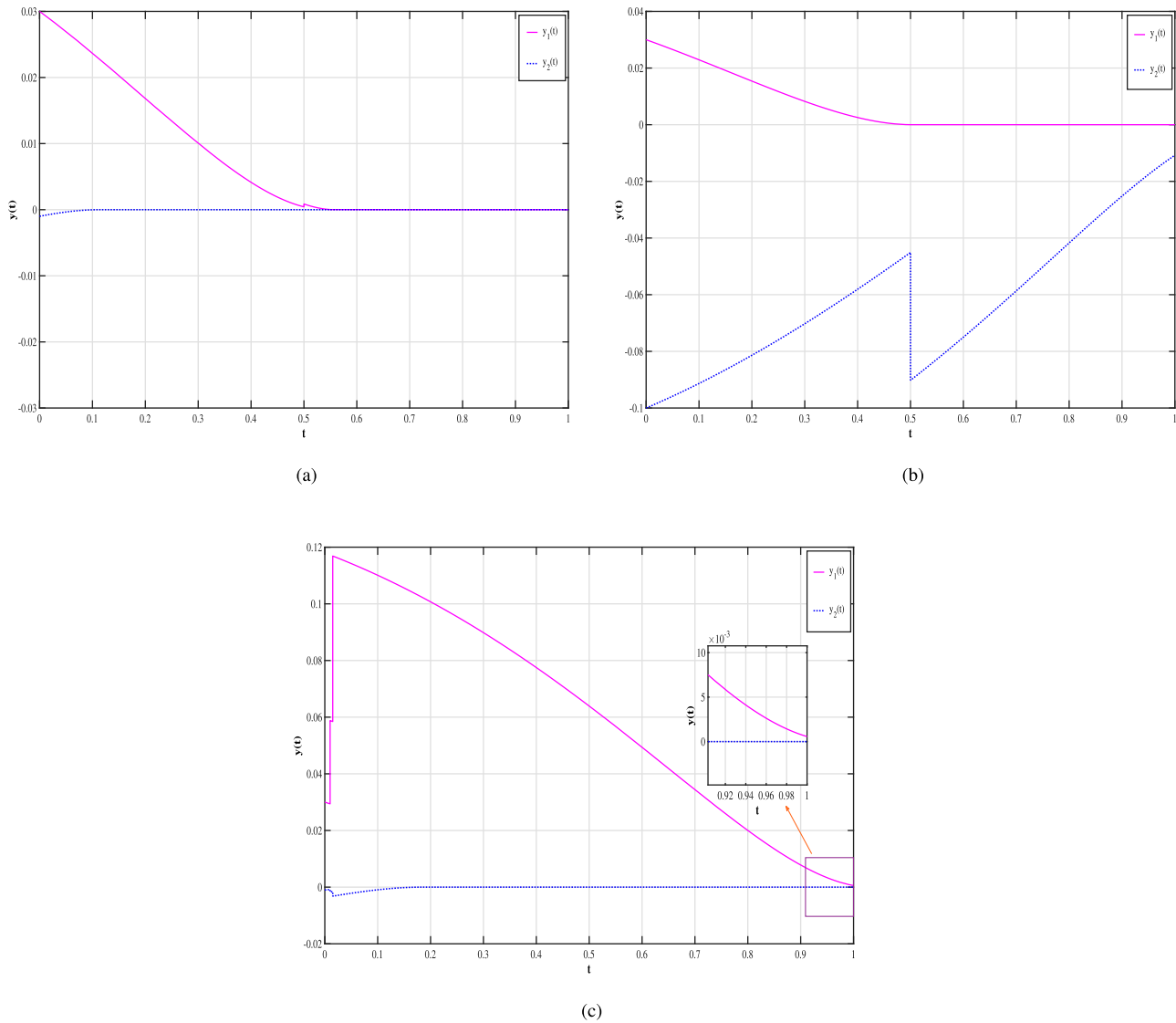


Figure 3. Simulations of Example 5.2.

from Figure 3(c) that it is not stabilised in the prescribed time $T_L = 1$. This justifies the essence of the conditions imposed on the impulsive sequence as given in Theorem 3.2.

Example 5.3: Let us look into the two-dimensional time-varying affine system (25) along with

$$f_a(t, y) = \begin{pmatrix} y_1 | y_2 | \\ -(2 + \sin t) | y_1 | y_2 \end{pmatrix}, \quad f_b(t, y) = \begin{pmatrix} y_1^2 \\ y_2 \end{pmatrix},$$

$$g_a(t, y) = \begin{pmatrix} \frac{2e^t}{e^{T_L}-1} & 0 \\ 0 & \frac{2e^t}{e^{T_L}-1} \end{pmatrix}, \quad g_b(t, y) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Consider the Lyapunov function $V_L(t) = V_L(t, y(t)) = [\text{sgn}(y(t))]^T Q(t) y(t)$, where $Q(t) = \begin{pmatrix} 2+\sin t & 0 \\ 0 & 1 \end{pmatrix}$. Then $\chi(t, y) = [\text{sgn}(y)]^T Q(t) f_a = 0$, $\tau(t, y) = [\text{sgn}(y)]^T Q(t) g_a = \begin{pmatrix} (2 + \sin t) \text{sgn}(y_1) \frac{2e^t}{e^{T_L}-1} & \text{sgn}(y_2) \frac{2e^t}{e^{T_L}-1} \end{pmatrix}$. Further, consider the disturbance as $\delta(t, y) = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$. Then $\tau(t, y) \delta(t, y) = \begin{pmatrix} (2 + \sin t) \text{sgn}(y_1) \frac{2e^t}{e^{T_L}-1} & \text{sgn}(y_2) \frac{2e^t}{e^{T_L}-1} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = (2 + \sin t) \frac{2e^t}{e^{T_L}-1} |y_1|$

$+ \frac{2e^t}{e^{T_L}-1} |y_2| = \{[\text{sgn}(y)]^T Q(t) y\} \zeta(t)$, with $\zeta(t) = \frac{2e^t}{e^{T_L}-1}$. So condition (i) of Theorem 4.1 is fulfilled. Also, conditions (ii) and (iii) are fulfilled with $\mu(t) = \frac{e^t}{e^{T_L}-1}$ and $l = 2$. Now, in accordance to Theorem 4.1 it can be verified that the system (25) is uniformly globally $P_S TS$ with respect to prescribed-time T_L over the impulsive sequence Λ_0 of Theorem 4.1 under the controllers

$$\Gamma_a(t, y) = - \begin{pmatrix} \frac{e^{T_L}-1}{2e^t(2+\sin t)} \text{sgn}(y_1) \\ \frac{e^{T_L}-1}{2e^t} \text{sgn}(y_2) \end{pmatrix}$$

$$\times \left\{ |y_1| \cos t + ((2 + \sin t) |y_1| + |y_2|)^p \right.$$

$$\times \left. (1 + (2 + \sin t) |y_1| + |y_2|)^{2-p} \frac{e^t}{e^{T_L}-1} \right\}$$

and

$$\Gamma_b(t, y) = - \begin{pmatrix} y_1^2 \\ y_2 \end{pmatrix}$$

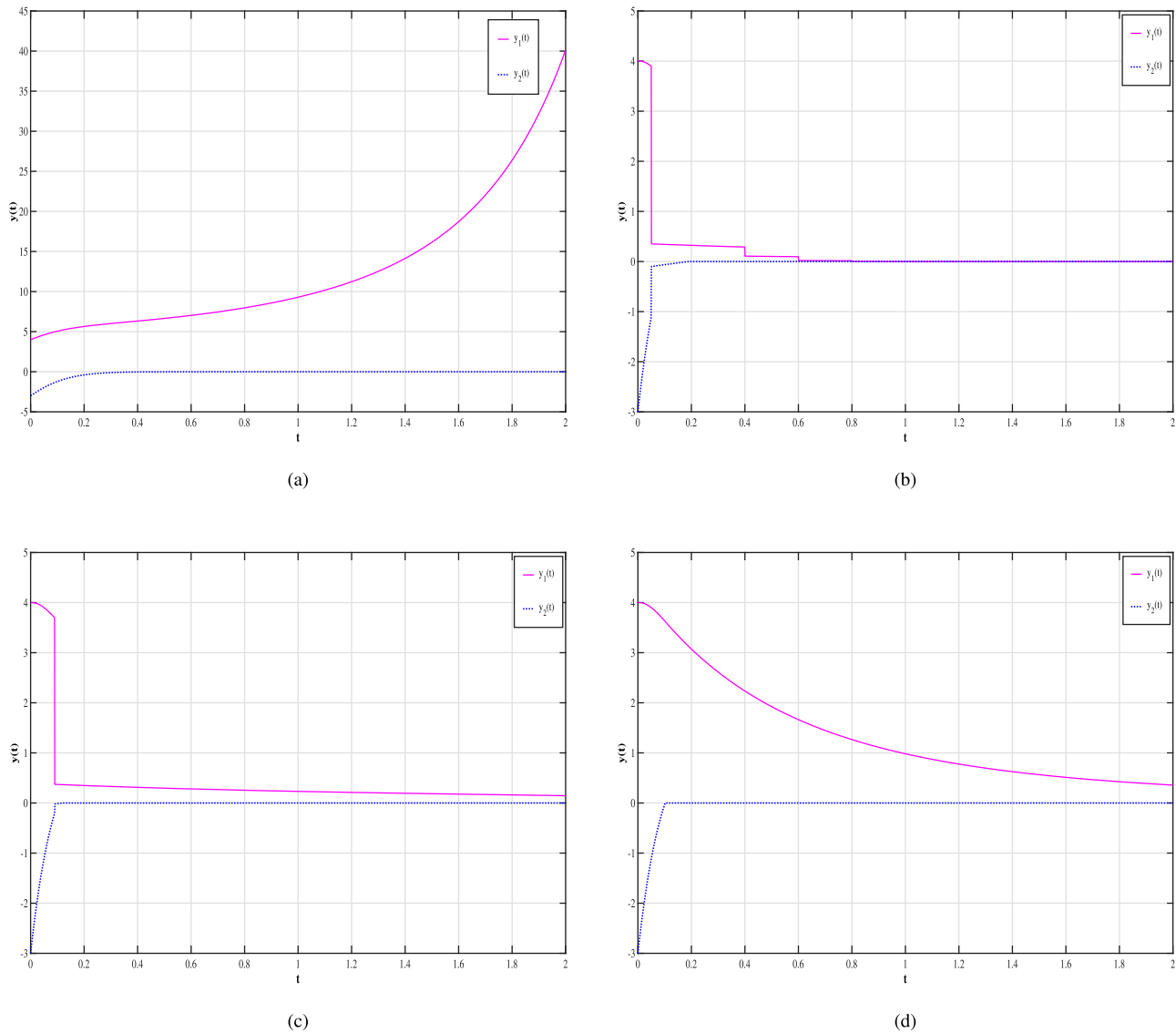


Figure 4. Simulations of Example 5.3.

$$+ \frac{p_1}{1 + (\alpha - p_1\alpha) ((2 + \sin t) |y_1| + |y_2|)} \times \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}.$$

For numerical simulation, let us consider $T_L^\diamond = 2, \alpha = 1, p = p_1 = \frac{1}{2}$. The trajectories of system (25) without any control law in the presence of disturbance term are given in Figure 4(a). Moreover, by simple calculation one can have $N \geq 5$ and $t_N \leq 1$. Further, consider the impulsive sequence $\{0.05, 0.4, 0.6, 0.8, 0.9\}$ and initial condition as $y_0 = (4, -3)^T$. From Figure 4(b) one can obtain that the states are stabilised to the origin within the prescribed time $T_L^\diamond = 2$. Moreover, if we consider $N = 1$, which goes against our condition, and choose impulsive point as $\{0.09\}$ one can have from Figure 4(c) that the states are not stabilised within time $T_L^\diamond = 2$. Similar sort of conclusion can also be obtained if we set $u_b = 0$, i.e. there is no impulsive control. Figure 4(d) validates this claim. These verifies the conclusion of Theorem 4.1.

6. Conclusion

The present work is mainly concerned with providing some adequate results for the P_S TS of time-dependent impulsive systems. Rigorously, with the help of time-varying gain functions and sequences of stabilising impulses, unstable continuous dynamics of the system are regulated first to reach to a stable neighbourhood of the origin and then to the origin in some desired prescribed time provided by the users. Further, some results are provided that accomplish P_S TS of impulsive systems under the influence of destabilising impulses within a local stable region around the origin. In addition, time-varying feedback controllers are also constructed for stabilising the time-varying affine systems within the prescribed time period. In addition, four scenarios are given to support the paper's objectives and suggested outcomes appropriately. In the future, it will be fascinating to provide such results corresponding to systems with delays in them. Further, building strategies for exact P_S TS will be concentrated on along with designing of event-triggered impulsive control mechanism, in the framework of Ai et al. (2024).

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Data availability

The authors declare that all the data generated and analysed during the study are present in the article.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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